



HELMHOLTZ AI

Introduction to Blablador and its applications

Blablador agenda

1. What is Blablador?
2. How is Blablador useful?
3. What can Blablador do?
4. Installing Blablador for VSCode
5. Other tools for agentic coding and chatting

What Is Blablador?

What is a language model provider?

What is a language model provider?



ChatGPT

What is a language model provider?



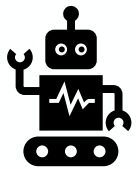
ChatGPT

OpenAI

What is a language model provider?



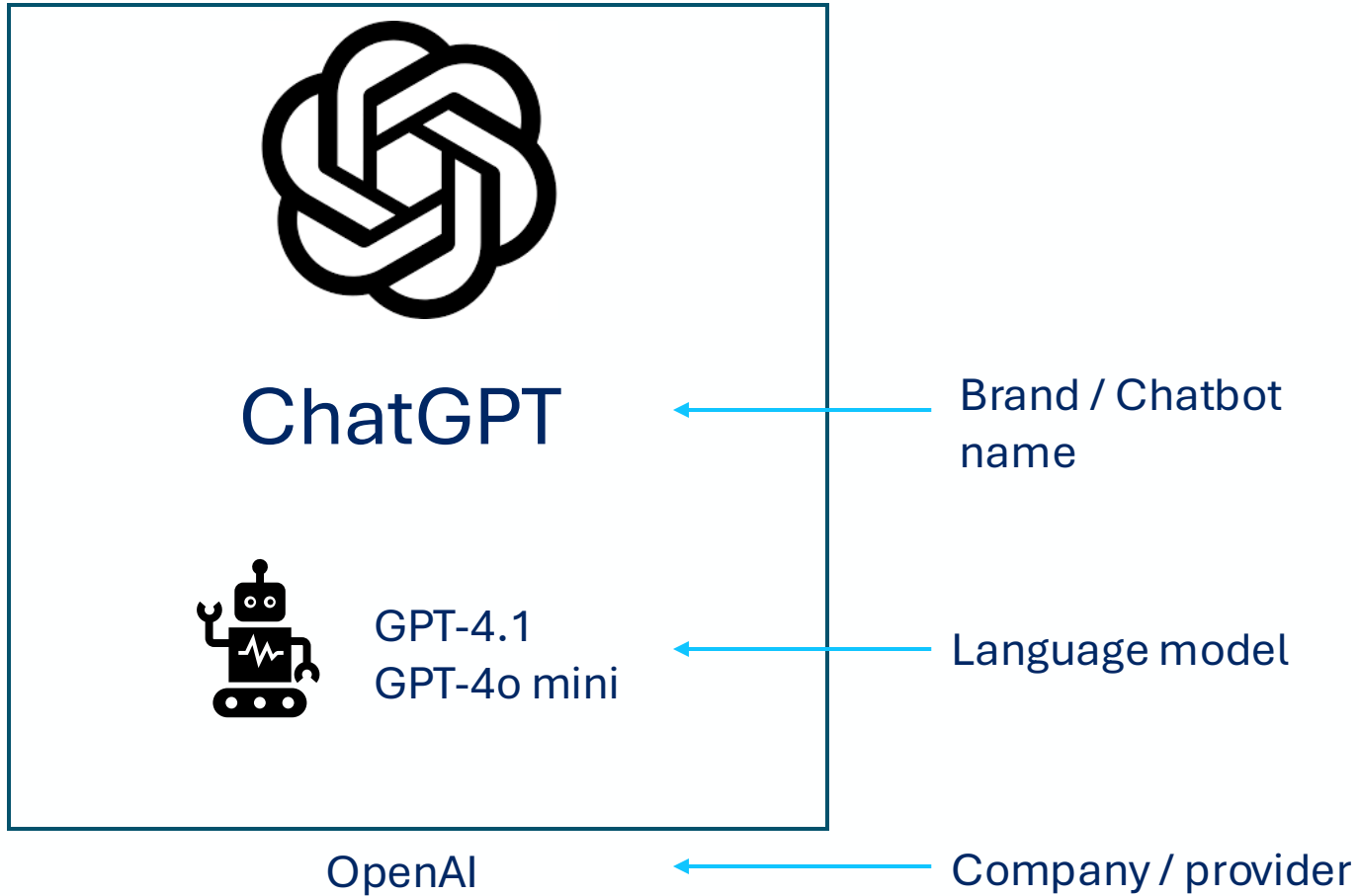
ChatGPT



GPT-4.1
GPT-4o mini

OpenAI

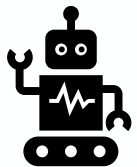
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ChatGPT

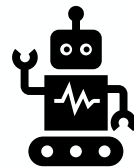


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OpenAI



Le Chat

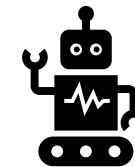


Mistral Large
Mistral Small

Mistral



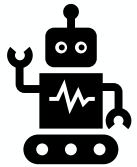
Claude



Claude Sonnet
Claude Opus

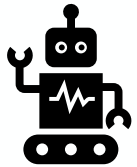
Anthropic

What do those models have in common?



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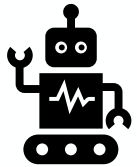
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- We cannot see what's inside.

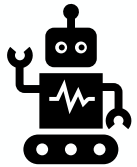
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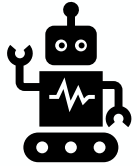
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- Require paid, professional plans when scaled.
- Data privacy concerns.
- A couple companies have the know-how.

Is there an alternative?



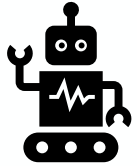
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	meta-llama/Llama-3.1-8B-Instruct
	Text Generation • 8B • Updated Sep 25, 2024 • 9.69M • 5.29k
	omni-research/Tarsier2-Recap-7b
	8B • Updated Aug 11, 2025 • 7.43M • 21
	openai/gpt-oss-20b
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	Qwen/Qwen2.5-14B-Instruct
	Text Generation • 15B • Updated Sep 25, 2024 • 4.9M • 300
	dphn/dolphin-2.9.1-yi-1.5-34b
	Text Generation • 34B • Updated Sep 8, 2025 • 4.06M • 54

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- We can see exactly what is going on inside.
- Free to download and run, for anyone.
- You make your own data policy.

But there's a catch

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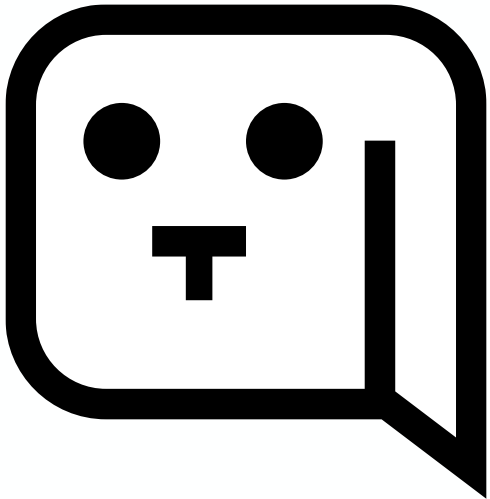
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- Most of these models are small (<30B parameters).
 - Larger models need a lot of hardware to even start.

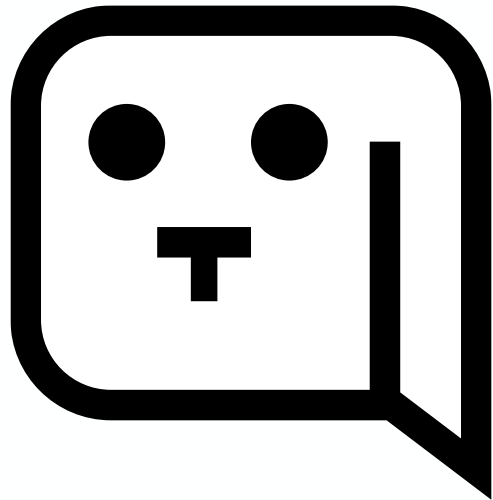
Meanwhile, the bigger the model is, the better.

Enter Blablador!



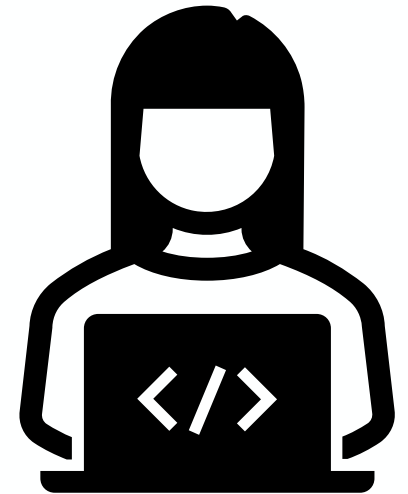
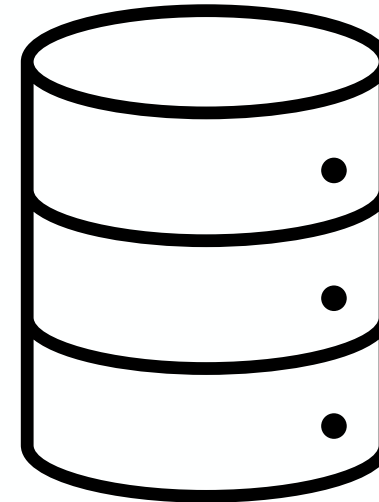
Blablador

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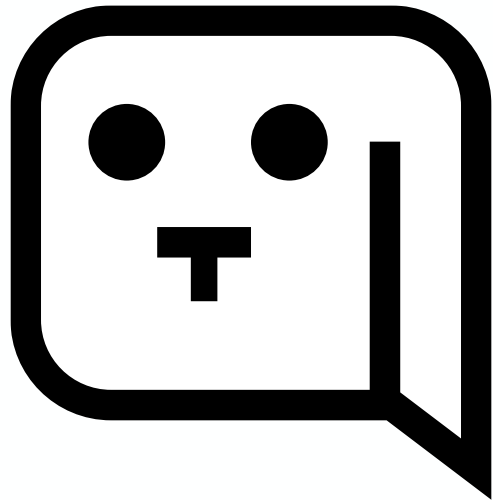
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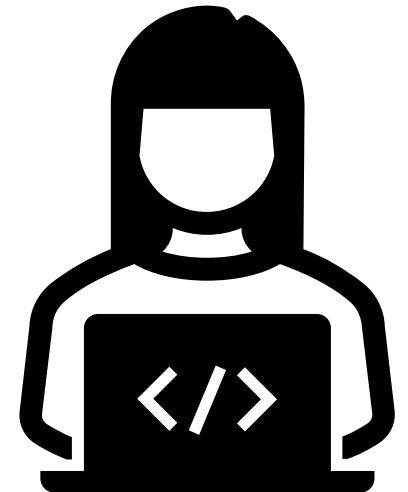
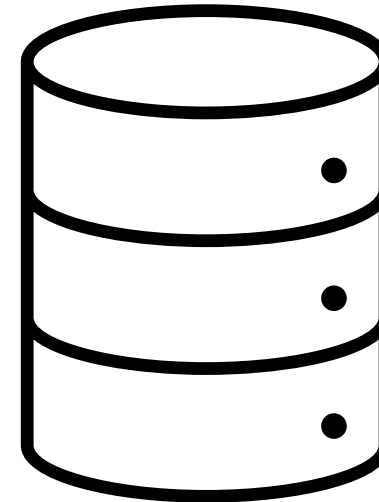
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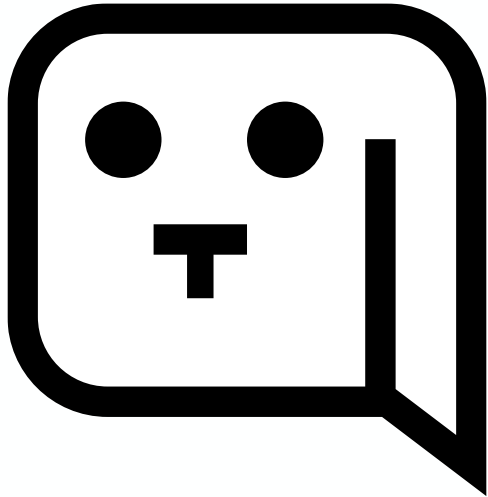
BLABLADOR is an evaluation server for Large Language Models (LLMs). Users can access a chat interface and an API [...]



Blablador

Blablador, the Helmholtz AI's Large Language Model (LLM) hosts both open-source models and those created by the Helmholtz Association.





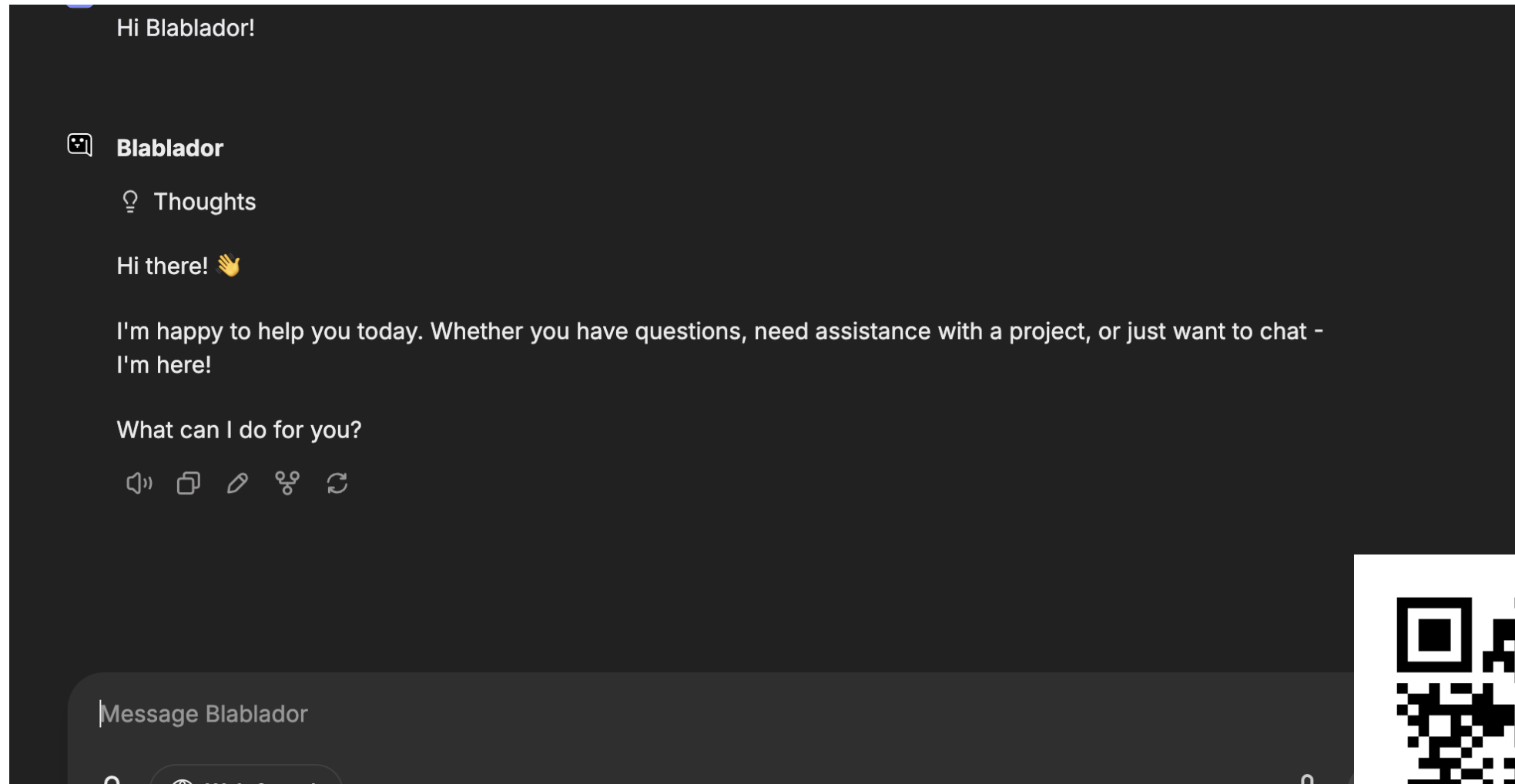
Blablador



ANTHROPIC



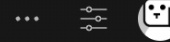
Mistral AI



helmholtz-blablador.fz-juelich.de/



0 - Ministral-3-14B-Instruct-2512 - The latest Ministral from Dec.2.2025 ▾ +



How are you doing?



0 - Ministral-3-14B-Instruct-2512 - The latest Ministral from Dec.2.2025

I'm just a computer program, so I don't have feelings, but I'm here and ready to help you with anything you need! 😊 How about you—how are *you* doing today? Anything on your mind or something I can assist with?



Follow up

Can you suggest some ways to boost my productivity today?

What's a good approach to managing stress if I'm feeling overwhelmed?

Do you have tips for improving my focus or work-life balance?

How can I stay motivated when working on long-term goals?

Are there any quick mood-boosting activities or habits you'd recommend?



Blablador supports OpenAI API for Tool Integration



LangChain



Some Blablador models support function calling for building Agentic AI

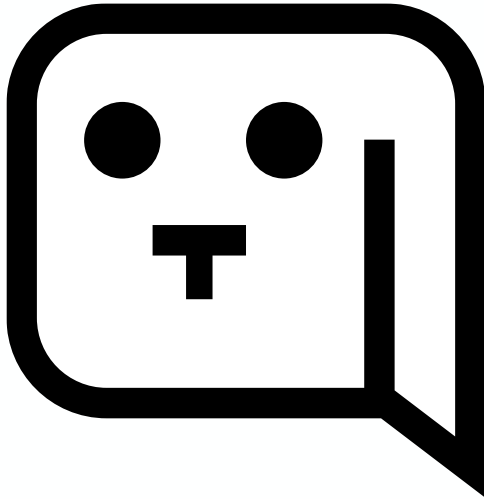


LangChain



Model Context Protocol

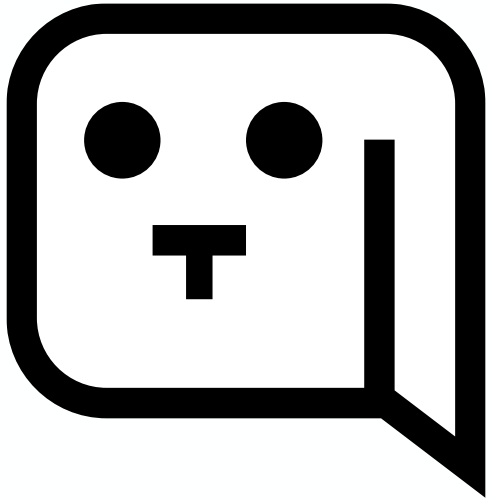
Blablador solves most of the issues with external providers



Blablador

- Hosted by the Helmholtz AI team @ FZJ
- Open-Source models
- No data stored at all.

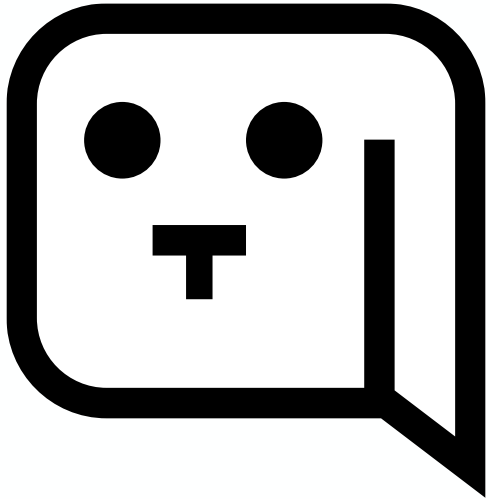
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- Free, unlimited usage within Helmholtz.

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- Open-Source models
- No data stored at all.
- We retain the know-how and control.
- Free, unlimited usage within Helmholtz.
- The best models are up to 1T parameters large.

Blablador in VSCode

Let's set up Blablador for VSCode

Get your access token at:

https://codebase.helmholtz.cloud/-/user_settings/personal_access_tokens

Get the config file from:

https://sdlaml.pages.jsc.fz-juelich.de/ai/guides/blablador_on_vscode/

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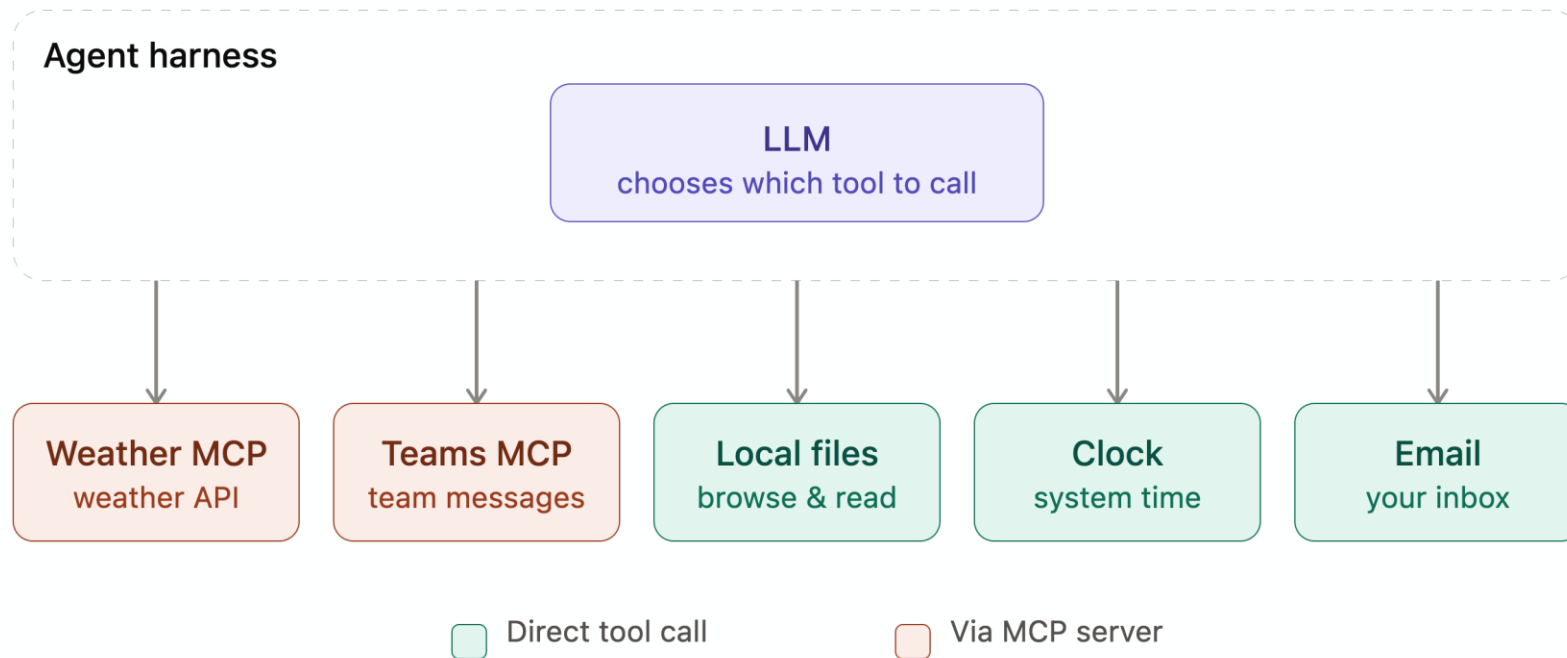
Get the config file from:

https://sdlaml.pages.jsc.fz-juelich.de/ai/guides/blablador_on_vscode/

Feel free to take a couple minutes to try that with your codebase!

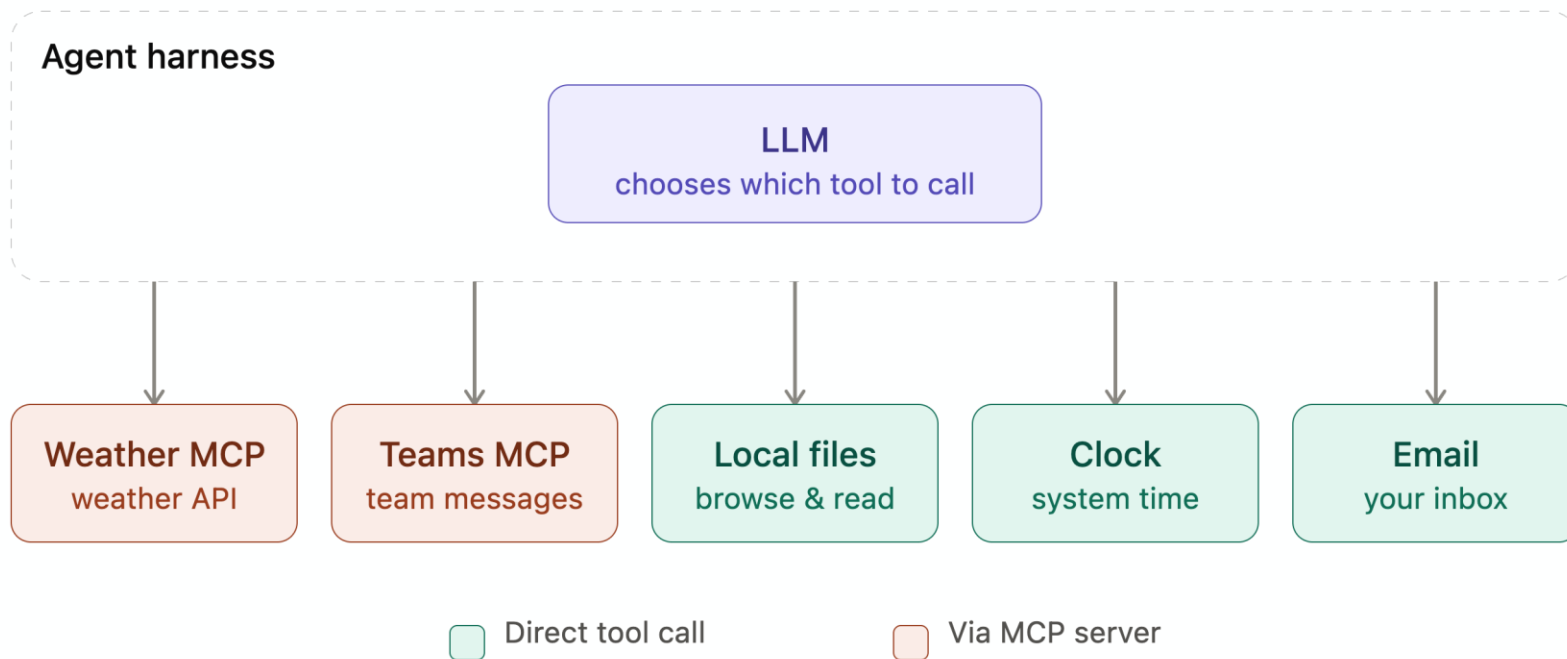
Agent harness

VSCode is one, but not the only one!



Agent harness

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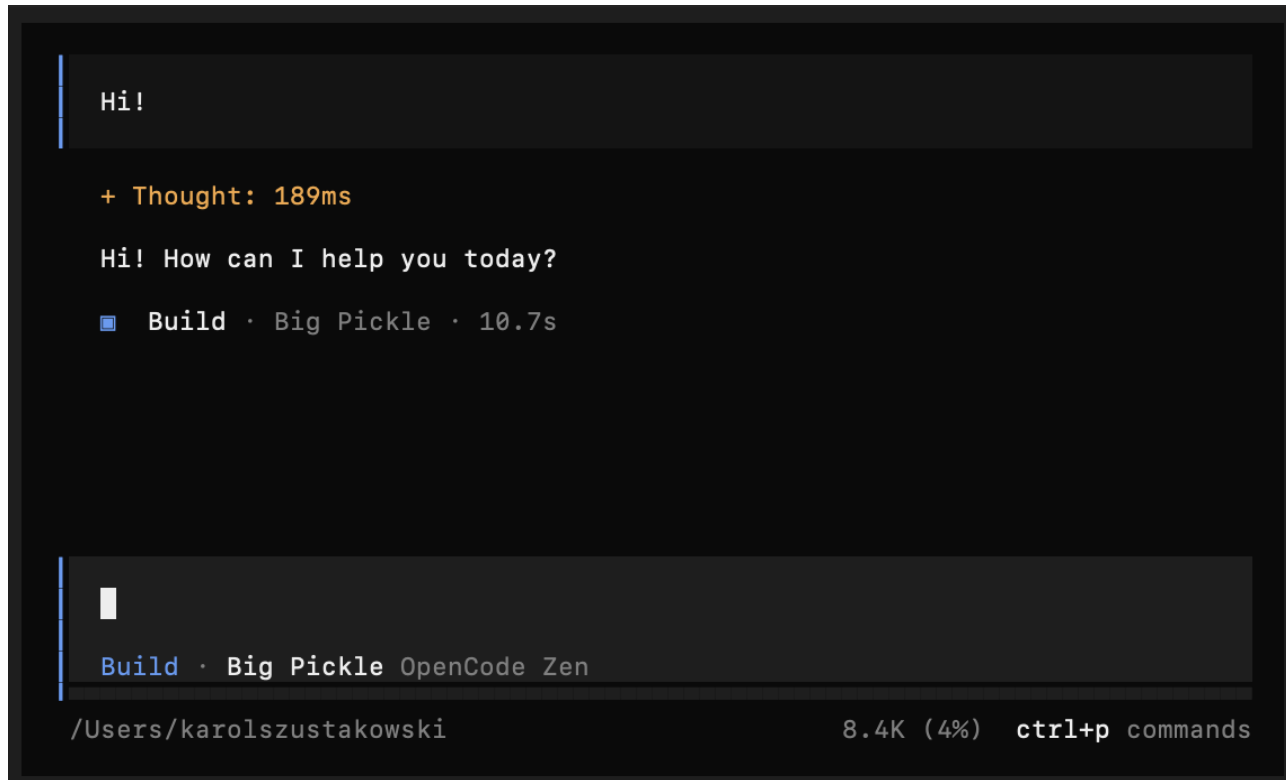


```

{
  "role": "assistant",
  "stop_reason": "tool_use",
  "content": [
    {
      "type": "text",
      "text": "Let me check the weather for that."
    },
    {
      "type": "tool_use",
      "id": "toolu_01xyz",
      "name": "get_weather",
      "input": {
        "location": "Munich, DE",
        "unit": "celsius"
      }
    }
  ]
}
  
```

Agent harness

OpenCode, Pi, the leaked Claude Code...



The screenshot shows a terminal window with a dark background. At the top, there is a text input field containing "Hi!". Below this, a system message is displayed: "+ Thought: 189ms". The AI's response is "Hi! How can I help you today?". Below the response, a status bar shows "Build · Big Pickle · 10.7s". At the bottom of the terminal, there is a text input field containing "Build · Big Pickle OpenCode Zen". The terminal's status bar at the very bottom shows the path "/Users/karolszustakowski", the file size "8.4K (4%)", and the command "ctrl+p commands".

```
Hi!
```

+ Thought: 189ms

Hi! How can I help you today?

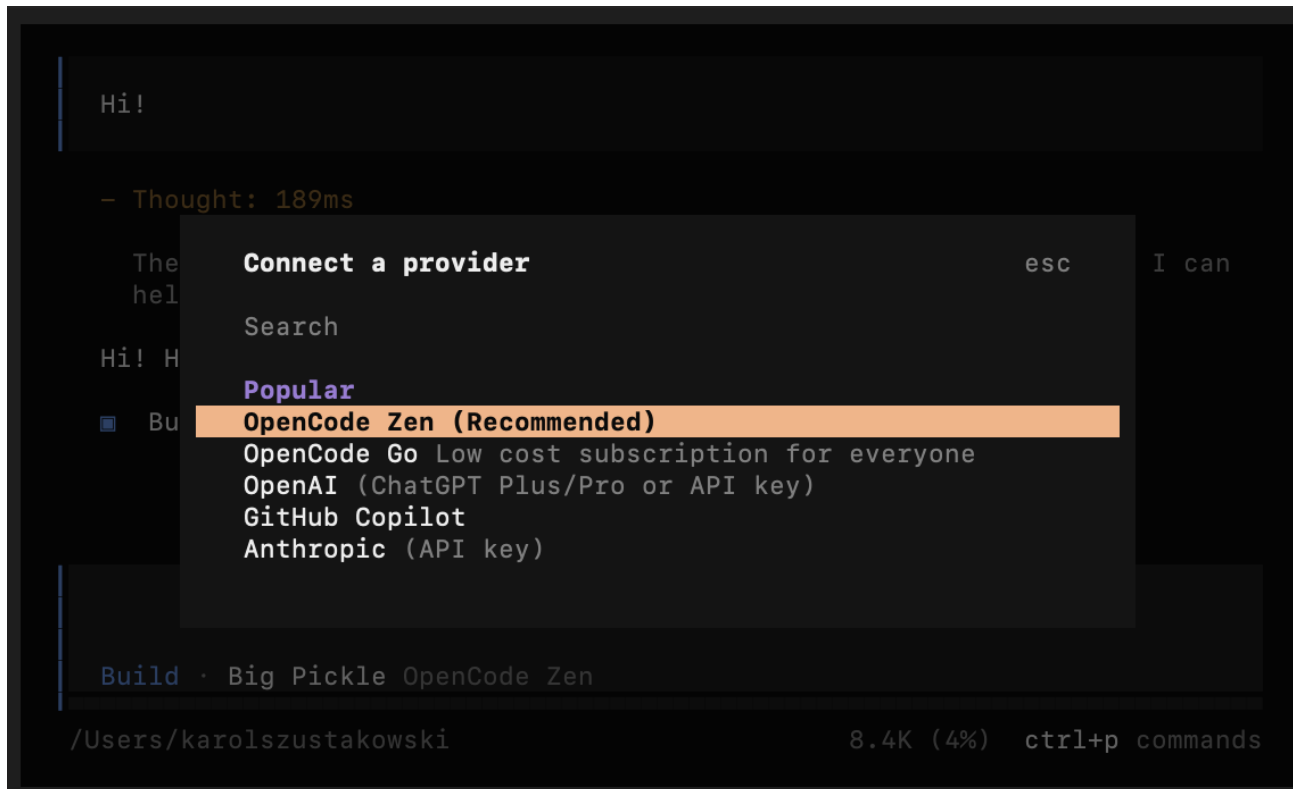
Build · Big Pickle · 10.7s

```
Build · Big Pickle OpenCode Zen
```

/Users/karolszustakowski 8.4K (4%) ctrl+p commands

Agent harness

OpenCode, Pi, the leaked Claude Code...



OpenCode 1.18.30 might not
Work with custom providers.

```
curl -fsSL
https://opencode.ai/install
| bash
```

Agent harness

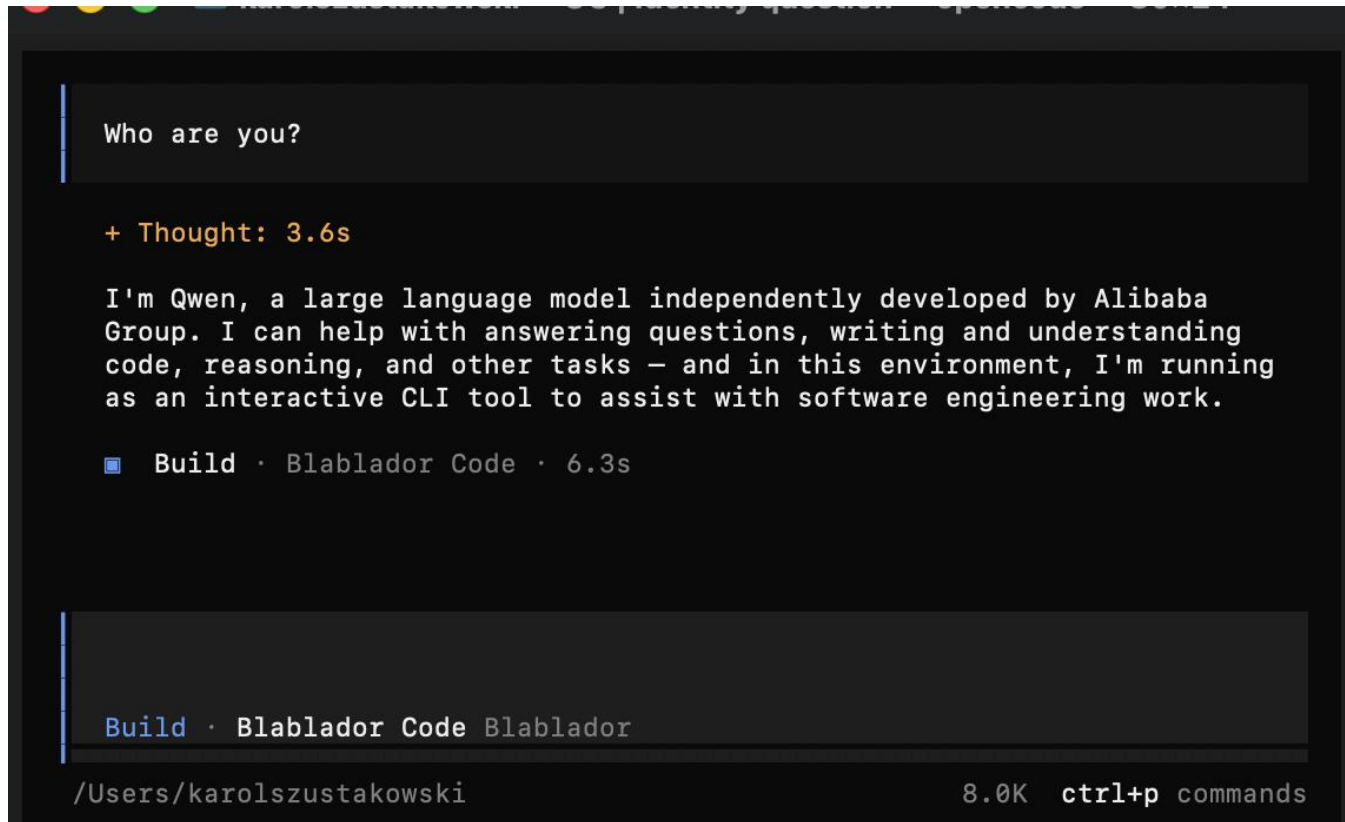
```
~/.config/opencode/opencode.jsonc
{
  "$schema": "https://opencode.ai/config.json",

  "provider": {
    "blablador": {
      "npm": "@ai-sdk/openai-compatible",
      "name": "Blablador",
      "options": {
        "baseURL": "https://api.blablador.fz-juelich.de/v1"
      },
      "models": {
        "alias-code": {
          "name": "Blablador Code"
        }
      }
    }
  },

  "model": "blablador/alias-code"
}
```

Agent harness

OpenCode, Pi, the leaked Claude Code...



The screenshot shows a terminal window with a dark background. At the top, a prompt box contains the text "Who are you?". Below this, a response from "Qwen" is displayed, preceded by a "+ Thought: 3.6s" label. The response text is: "I'm Qwen, a large language model independently developed by Alibaba Group. I can help with answering questions, writing and understanding code, reasoning, and other tasks – and in this environment, I'm running as an interactive CLI tool to assist with software engineering work." Below the response, a status bar shows "Build · Blablador Code · 6.3s". At the bottom of the terminal, there is a prompt "Build · Blablador Code Blablador" and a footer showing the path "/Users/karolszustakowski" and the text "8.0K ctrl+p commands".

```
Who are you?
```

+ Thought: 3.6s

I'm Qwen, a large language model independently developed by Alibaba Group. I can help with answering questions, writing and understanding code, reasoning, and other tasks – and in this environment, I'm running as an interactive CLI tool to assist with software engineering work.

Build · Blablador Code · 6.3s

Build · Blablador Code Blablador

/Users/karolszustakowski 8.0K ctrl+p commands

Bonus: Which LLM is the best?

LLM Evaluation is inherently very hard, and data leakages are commonplace

Bonus: Which LLM is the best?

LLM Evaluation is inherently very hard, and data leakages are commonplace

Often, the only real evaluation is by comparison

Bonus: Which LLM is the best?

<https://arena.ai/> - let's give it a shot

Arena Overview Text Code Vision Document Text-to-Image Image Edit Search Text-to-Video Image-to-Video Start Voting

New Chat
Leaderboard
Search

Leaderboard Overview

See how leading AI models stack up across text, image, vision, and more. This page provides a high-level snapshot of each Arena. Explore dedicated tabs for deeper insights. [Learn more here.](#)

Text 3 days ago

Rank	Model	Score	Votes
1	AI claude-opus-4-6	1504	9,170
2	AI claude-opus-4-6-thinki...	1502	8,313
3	G gemini-3.1-pro-preview	1500	4,041
4	XI grok-4.20-beta1	1491	5,280
5	G gemini-3-pro	1485	39,923

Code 5 days ago

Rank	Model	Score	Votes
1	AI claude-opus-4-6-thinki...	1556	2,553
2	AI claude-opus-4-6	1555	3,273
3	AI claude-sonnet-4-6	1523	3,150
4	AI claude-opus-4-5-202511...	1497	11,859
5	AI claude-opus-4-5-202511...	1475	11,994

Join us on YouTube

Survey

Scan the QR code or follow the link: [LLMs in Practice: Insights from User and Developer Perspective - 2026 @ Helmholtz Munich – Fill in form](#)

LLMs in Practice: Insights from
User and Developer Perspective -
2026 @ Helmholtz Munich



Helmholtz AI Consultants

Who are we and what is our mission?

What is our mission?

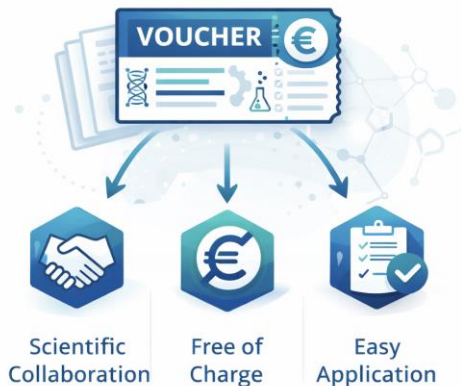


- ❖ Maximise research impact by democratizing access to AI

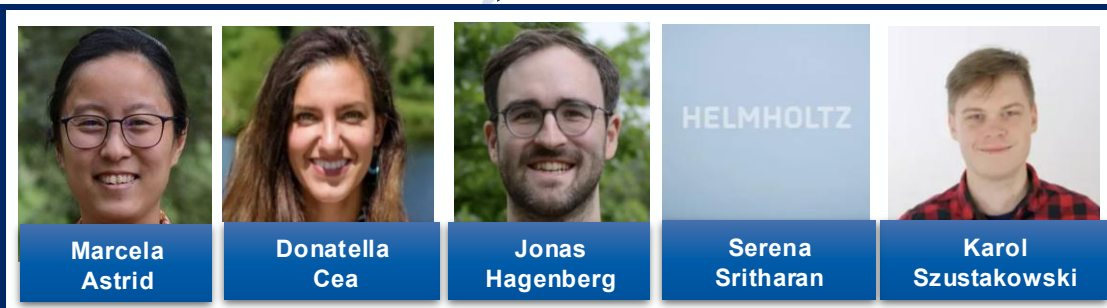


- ❖ We empower health researchers with tailored AI solutions and strive to translate them into reusable tools for the community.

How do we do that?



<https://www.helmholtz.ai/you-helmholtz-ai/ai-consulting/>



Agenda – Day 2

- 09:00 - 09:15 Welcome & Group activity
- 09:15 - 10:15 Attention, self-attention, transformer architecture
- 10:15 - 10:30 Pre-training vs fine-tuning and foundation models (intro to the hands-on session)
- 10:30 - 10:45 Break
- 10:45 - 12:15 Hands-on: fine-tune a model on a downstream task
- 12:15 - 12:30 Q&A, Wrap-up, and Conclusion

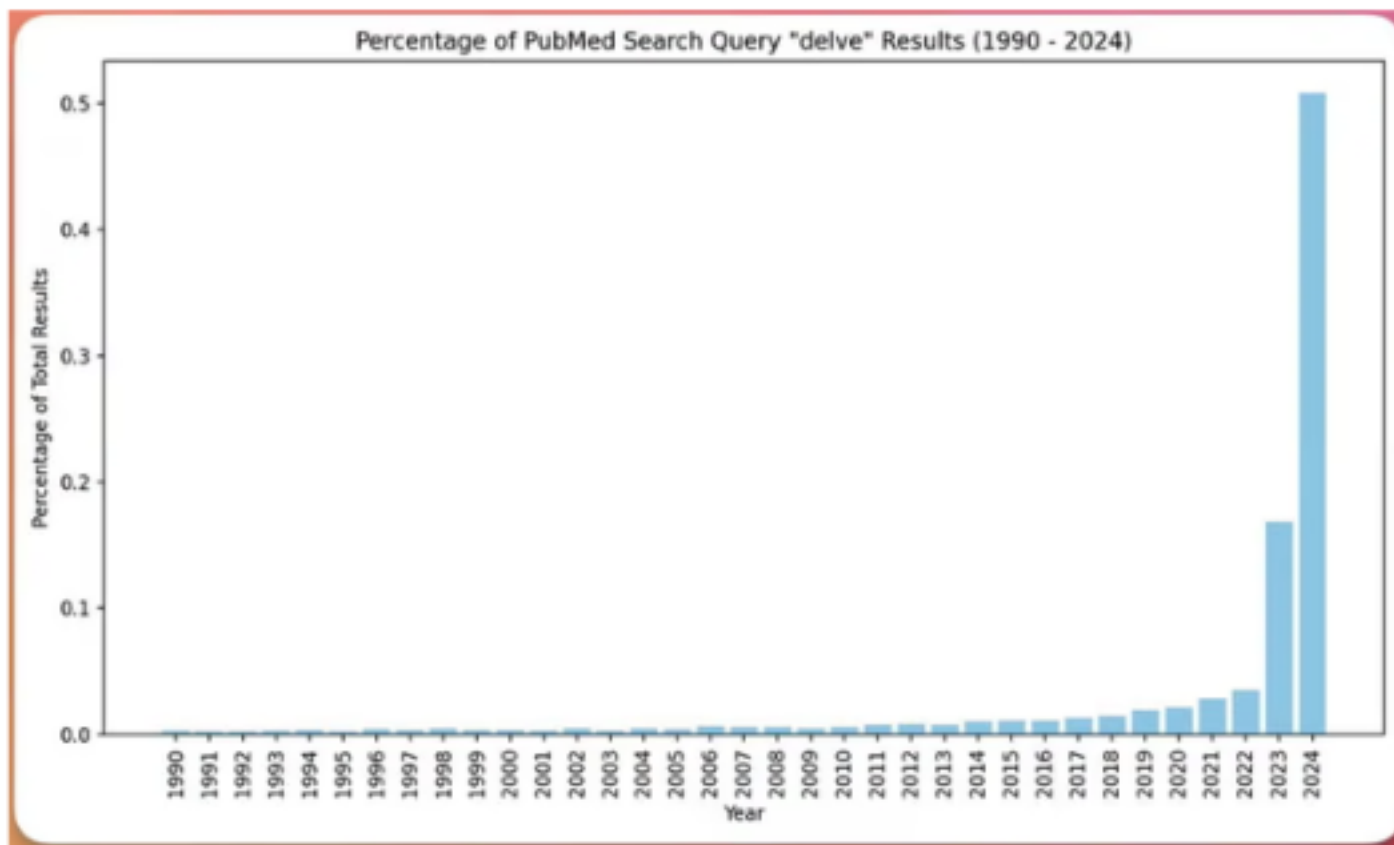
Mixed-up Sentence

- ❑ You will be split into small breakout groups (3–4 people)
- ❑ Each group receives 7 statements about LLMs
- ❑ Your task:
 - ❑ Identify **the one statement that does not fit with the others**
 - ❑ Be ready to **briefly explain your reasoning**
- ❑ **All statements are correct; one is superfluous**, i.e. it is true but was not discussed during this session
- ❑ **Time:** 8–10 minutes

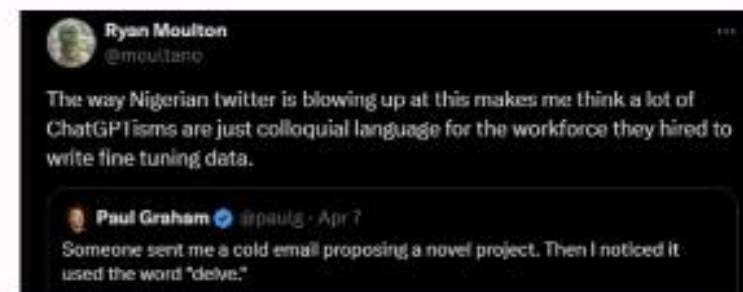
Mixed-up Sentence

1. The knowledge acquired during training is embedded in the model parameters and used at inference time, rather than retrieved from the original dataset.
2. Identical inputs do not always lead to identical outputs when interacting with a language model.
3. The behavior of a language model may be influenced by provider-defined instructions that are not visible to the user.
4. Biases and behavioral patterns can be introduced or amplified not only by the training data, but also during post-training, for example through human feedback.
5. Language models generate text incrementally by selecting likely continuations given the context.
6. The use of language models does not remove the researcher's responsibility for the validity of the results.
7. Increasing model size typically comes with higher computational and infrastructure demands.

Bias doesn't only come from training data



It was suggested that this may have something to do with the fact that OpenAI used Kenyan workers for fine-tuning the model through RLHF. I am not sure I believe this: Kenya and Nigeria are quite far and different. I admit that I have no idea if formal English in Kenya has similarities with formal English in Nigeria.



Honestly, that's not important. It doesn't really matter whether a word was overrepresented in the training data, or received high scores during reinforcement learning from human feedback. The essential point is that ChatGPT is designed to emulate human writing as closely as possible, so its quirks ² almost always come from human writing.

Introduction to Transformers

Flipped Classroom Approach

We will follow this material:

https://xai-tutorials.readthedocs.io/en/latest/_ml_basics/transformer.html#

Using a sort of **Flipped Classroom** approach in order to:

- Reduce Zoom fatigue
- Support different learning paces
- Encourage active learning
- Provide reusable material

Beyond Prompting: Modern AI Applications

Model Context Protocol (MCP)

Curious to learn how to design and implement them to build agentic systems powered by modern LLMs?

Join our in-person workshop!
When? Next Friday, 25 of September
Where? At Neuherberg Campus

How?

[From MCP Server to Agent: A Hands-On Workshop on Building Tool-Using LLM Systems | CaMS](#)

Survey

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